

PROJECT PROPOSAL
GROUP PROJECT CST 394-2

Group No: CST 06

PyramidEdu

Smart Institute Management and Student Performance
Prediction System

Computer Science and Technology
Department of Computer Science and Informatics
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DECLARATION

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Mr.U.E.Ranasooriya [umesh@uwu.ac.lk] and co supervisor Ms.J.A.N. Navodya [navo.jayakodi@gmail.com]. This Project Proposal is submitted for the partial fulfillment of the requirement of the course unit CST 394-2, Project II for the degree of Computer Science and Technology.

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1. Introduction

1.1. Project Title

PyramidEdu: Smart Institute Management and Student Performance Prediction System

1.2. Project Description

The PyramidEdu system is a centralized web and mobile application designed to support the academic and administrative operations of modern educational institutes. It is developed to address common inefficiencies in small to medium-scale education centers such as manual record keeping, fragmented data management, poor communication, and limited access to real-time academic insights [\[1\]](#).

PyramidEdu integrates modern technologies and intelligent features to streamline institutional processes, including student management, subject enrollment, QR-based real-time attendance tracking, examinations and quizzes, and centralized financial operations. Students can enroll in multiple subjects, and fees are automatically calculated based on selected subjects.

The system includes a mobile application that allows students to access attendance records, academic progress, subject details, examination updates, and notifications from anywhere. An NLP-powered AI chatbot (available to Students and Teachers only) provides intelligent academic assistance, PDF summarization, AI question generation, and personalized study recommendations using the Google Gemini API [\[6\]](#). The system also introduces intelligent student performance prediction and an exam recommendation system. Automated student progress reports are generated and sent to parents to improve communication and transparency [\[2\]](#).

Role-based dashboards for Admin, Manager, and Teachers ensure efficient management of institutional activities. The system supports financial control by restricting access for students with unpaid fees and includes a reward mechanism where students earn progress points and high performers receive incentives.

1.3. Background and Motivation

Effective management of educational institutions is essential for smooth academic operations, accurate record keeping, and improved student performance. Many small to medium-scale institutes still rely on manual or partially digitized systems, leading to inefficiencies such as inconsistent student tracking, delayed communication, data redundancy, and lack of centralized management. Studies have shown that digitizing institutional processes significantly improves administrative efficiency and reduces error rates in educational environments.

The growing adoption of AI and analytics in education has opened new opportunities for student performance monitoring and personalized learning. Research by Hussain et al. [3] demonstrated that analyzing student attendance, quiz performance, and assignment scores enables early identification of at-risk students with high accuracy. Similarly, Kabakchieva [4] showed that trend-based classification of academic records can effectively identify underperforming students early in the academic cycle, enabling timely interventions that improve pass rates.

Therefore, there is a strong need for a centralized and intelligent system that improves efficiency, enhances communication, and supports data-driven decision-making. This project, PyramidEdu, is developed to address these issues by integrating academic, administrative, and financial processes into a single smart platform.

1.4. Problem in Brief

Effective management of educational institutions is essential for smooth academic operations, financial control, and improved student performance. However, many small to medium-scale institutes still rely on fragmented or partially digitized systems, leading to several operational challenges.

- **Lack of Centralized System**

Academic, attendance, and financial processes are managed using separate systems, resulting in inconsistent and unorganized data management.

- **Weak Fee Management Control**

There is no strict automated mechanism to restrict student access when fees are unpaid for a certain period (e.g., 3 months), causing financial inefficiencies.

- **Limited Subject Flexibility**

Students cannot dynamically select or unselect subjects after enrollment, and subject-wise fee calculation is not properly automated.

- **Poor Performance Tracking**

Institutions lack continuous monitoring of student performance using attendance, marks, and assessments.

- **No Intelligent Prediction System**

Existing systems do not predict student performance or recommend suitable academic improvements.

- **Lack of Incentive Mechanism**

There are no structured reward systems such as progress points or incentives for high-performing students.

- **Parent Communication Gap**

There is no automated system to regularly update parents about student attendance, performance, and progress.

These issues lead to inefficiencies in administration, poor transparency, weak financial control, and limited data-driven decision-making in educational institutions.

1.5. Proposed Solution

We propose developing **PyramidEdu: Smart Institute Management and Student Performance Prediction System**, an advanced centralized educational management platform that transforms traditional academic and administrative operations into a unified, intelligent digital ecosystem. PyramidEdu is designed to address key challenges such as manual record keeping, fragmented systems, poor communication, and lack of data-driven decision-making by integrating automation, QR technology, and Artificial Intelligence.

The PyramidEdu platform provides a web and mobile-based environment where administrators, teachers, students, and parents can interact seamlessly. By centralizing academic, financial, and communication processes into a single system, the platform improves efficiency, transparency, and institutional coordination.

Key features such as QR-based attendance tracking, AI-powered performance prediction, automated fee control, intelligent exam recommendation, and chatbot-based academic assistance collectively ensure a smart and responsive educational experience. These features enable institutions to monitor student performance in real time while supporting personalized learning and early intervention strategies.

1. PyramidEdu will provide a **centralized management system**, integrating academic records, attendance, examinations, and financial operations into a single platform to eliminate data fragmentation and improve operational efficiency.
2. The system will implement a **QR-based Attendance System**, enabling fast, accurate, and real-time attendance tracking, reducing manual errors, and improving monitoring reliability.
3. A **Dynamic Subject Management System** will allow students to enroll in multiple subjects and modify selections, with automatic subject-wise fee calculation and updates.
4. The platform will introduce an **Automated Fee Control Mechanism**, which restricts system access for students with unpaid fees beyond a defined period, ensuring financial discipline within the institute.

5. An **AI-Based Student Performance Prediction module** will detect
Detecting at-risk students at an early stage using a Weighted Scoring and
Trend Analysis engine [5] that evaluates attendance, marks, quiz results,
and academic growth patterns to generate AI-assisted intervention alerts.
6. A **Reward and Incentive System** will motivate students by assigning
progress points based on academic performance and attendance,
encouraging continuous improvement.
7. The system will include an **AI Chatbot Assistant**, providing an NLP-
powered AI chatbot for Students and Teachers with free-text query support,
PDF summarization, AI question generation, academic recommendations,
and personalized study guidance via the Google Gemini API [6].
8. An **Automated Parent Communication System** will generate and send
regular student progress reports, ensuring transparency and strengthening
parent engagement in student development.

By combining intelligent automation, predictive analytics, and centralized data management, PyramidEdu enhances institutional efficiency, improves decision-making, and supports a scalable digital education ecosystem for future expansion.

1.6. Project Aim and Objective

1.6.1 Aim

The aim of this project is to develop PyramidEdu, a comprehensive and intelligent Institute Management System that enhances academic and administrative operations through centralized management, AI-driven automation, and real-time communication. The system is designed to streamline student management, attendance tracking, subject and fee management, academic monitoring, and parent communication while providing AI-assisted performance prediction, exam recommendation, and chatbot assistance for students and teachers.

1.6.2 Objectives

I. Enable a Centralized and Scalable Institute Management System:

To integrate academic, attendance, financial, subject management, communication, reporting, and monitoring functionalities into a single centralized platform while ensuring scalability, modularity, and support for future system enhancements and integrations.

II. Implement Smart Attendance, Subject, and Financial Management:

To provide QR-based real-time attendance tracking, dynamic subject selection and unselection, automated subject-wise fee calculation, and automated fee restriction mechanisms for efficient academic and financial management.

III. Integrate AI-Assisted Academic Support and Performance Analysis:

To implement an AI-assisted student performance prediction system using Weighted Scoring and Academic Growth Trend Analysis [\[5\]](#) on attendance, marks, and assessment data, combined with intelligent exam recommendations through performance-tier analysis and chatbot assistance via the Google Gemini API [\[6\]](#) to improve academic decision-making and personalized learning.

IV. Enhance Student Engagement and Parent Communication:

To encourage academic motivation through progress points and reward mechanisms while automatically providing parents with student progress reports and academic updates for improved transparency and engagement.

V. Deliver a Secure and User-Friendly Web and Mobile Experience:

To develop responsive web and mobile applications with secure role-based access control, efficient navigation, and reliable accessibility for students, teachers, and administrators across multiple devices.

VI. Support Institutional Monitoring and Decision-Making:

To provide centralized dashboards, analytics, and automated reporting features that assist Administrators and teachers in effectively monitoring institutional operations and academic performance.

2. Literature Review

This section reviews existing research related to the key components of PyramidEdu: educational management systems, QR-based attendance tracking, AI-assisted student performance prediction and analytics, chatbot integration, and automated parent communication systems.

2.1 Educational Management Systems

Almuhaisen and Al-Adwan [\[1\]](#) examined the adoption of digital management systems in higher education institutions and found that centralized platforms significantly reduce administrative overhead, improve data accuracy, and enhance institutional transparency. Their study highlighted those institutions still relying on paper-based or fragmented tools to suffer from data inconsistency and delayed communication, which are precisely the challenges of PyramidEdu targets.

Romero and Ventura [\[2\]](#) conducted a comprehensive survey of educational data mining and learning analytics, establishing that integrated digital platforms enable data-driven insights into student behavior and academic outcomes. Their work underpins PyramidEdu's approach to combining management features with predictive analytics into a single system.

2.2 QR Code-Based Attendance Systems

Ahmed et al. [\[7\]](#) proposed a QR code-based classroom attendance system and demonstrated that it reduces manual data entry time by up to 78% compared to traditional roll calls. Their server-side QR validation design directly influenced PyramidEdu's attendance module.

Wei, X., Manori. [\[8\]](#) extended QR-based attendance to mobile platforms, showing that integrating QR scanning within a mobile application increases student compliance and reduces proxy attendance by over 90%. PyramidEdu adopts this mobile-integrated approach via its React Native application.

2.3 AI-Assisted Student Performance Prediction and Analytics

Hussain et al. [3] analyzed student academic records comprising attendance, quiz scores, and assignment marks and confirmed that combining multiple academic indicators produces more reliable early-warning signals than any single metric alone. This finding directly informs PyramidEdu's prediction approach, which uses a Weighted Scoring formula applied across four key indicators: $\text{Performance Score} = (\text{Attendance} \times 0.30) + (\text{Quiz Average} \times 0.20) + (\text{Assignment Average} \times 0.20) + (\text{Exam Average} \times 0.30)$. The resulting score is then mapped to one of four performance categories: Excellent (80–100), Good (60–79), Average (40–59), or At-Risk (below 40). This rule-based scoring approach is implemented entirely within the Node.js backend using TypeScript, requiring no external machine learning library or pre-trained model.

Baker and Inventado [5] established that rule-based academic analytics and weighted scoring approaches are highly practical for student performance monitoring in institutional settings, particularly when labeled training datasets are limited or unavailable. Their work supports PyramidEdu's design of an Academic Growth Trend Analysis module that compares a student's score across sequential assessments to detect trajectory patterns. If quiz scores follow a consistently declining path (e.g., $75 \rightarrow 65 \rightarrow 52 \rightarrow 40$), the system flags the student as Declining and triggers an automated alert to the teacher and parent. Conversely, a rising trajectory (e.g., $40 \rightarrow 52 \rightarrow 65 \rightarrow 75$) marks the student as Improving. A flat trajectory within a ± 5 point range is classified as Stable. This deterministic trend engine classifies academic growth without requiring any pre-trained machine learning model.

Kabakchieva [4] demonstrated that early identification of declining academic trends enables timely interventions that significantly improve student pass rates. PyramidEdu incorporates this principle by connecting its Weighted Scoring and Trend Analysis engine to the Google Gemini API [6]. When a student is classified as At-Risk or Declining, the backend constructs a structured prompt containing the student's attendance percentage, quiz average, assignment average, and exam average, and submits it to Gemini, which returns a short, personalized academic recommendation. This AI-generated recommendation is then displayed on the student dashboard and included in the automated parent report, providing actionable guidance without requiring complex model training or large historical datasets.

2.4 AI Chatbot Integration in Education

Winkler and Sollner [\[9\]](#) systematically reviewed chatbot deployments in educational settings and found that AI-powered chatbots capable of understanding free-text natural language input significantly improve student engagement and academic support outcomes. This finding directly informs PyramidEdu's decision to implement a Natural Language Processing (NLP) chatbot using the Google Gemini API [\[6\]](#), enabling Students and Teachers to type questions freely and receive intelligent, context-aware academic responses.

Clarizia et al. [\[10\]](#) demonstrated that integrating a university support chatbot with live academic databases reduces advisor workload by 40% and improves student query resolution time from an average of 48 hours to under 2 minutes. PyramidEdu's chatbot is similarly connected to live student data via the Gemini API [\[6\]](#), enabling context-aware responses about marks, attendance, and fee status.

2.5 Automated Parent Communication Systems

Research in educational communication consistently demonstrates that regular, structured reporting to parents significantly improves student academic outcomes and engagement. Studies show that when parents receive timely updates on attendance, assessment scores, and academic progress, students exhibit higher motivation, lower absenteeism, and improved completion rates. This evidence supports the inclusion of an Automated Parent Communication System in PyramidEdu, which periodically generates and delivers structured student progress reports to parents through digital channels such as email and in-app notifications [\[2\]](#).

Romero and Ventura [\[2\]](#) further established that data-driven communication between institutions and families, enabled by integrated digital platforms, enhances transparency and enables guardians to take informed action in supporting their child's academic journey. PyramidEdu's parent reporting feature leverages this principle by automatically generating concise summaries of each student's attendance percentage, performance score, trend category, and Gemini AI-generated recommendations, ensuring parents receive meaningful and actionable information rather than raw data.

2.6 Summary

The reviewed literature confirms the effectiveness of individual components that PyramidEdu integrates centralized educational management platforms, QR-based attendance systems, AI-assisted rule-based performance prediction and trend analytics, AI chatbots, and parent communication systems. However, no existing system reviewed combines all these components into a single unified platform tailored for small to medium-scale private education centers in developing countries. PyramidEdu addresses this gap by delivering an integrated, cost-effective, and AI-augmented institutional management system deployable with standard web infrastructure, using a practical Weighted Scoring and Trend Analysis engine supported by Gemini AI recommendations rather than a complex pre-trained machine learning model.

3. Methodology

3.1. Agile Methodology

The PyramidEdu Smart Institute Management and Student Performance Prediction System is developed using Agile Methodology. This approach ensures a flexible, iterative, and collaborative development process, allowing the team to continuously improve the system based on stakeholder feedback and changing requirements. Agile methodology was specifically chosen because the project requires regular discussions with the client and continuous validation of system functionalities throughout development. Since institutional requirements, workflows, and feature expectations may evolve during the project lifecycle, Agile enables the team to adapt quickly to modifications without affecting overall system progress. By dividing the project into short development iterations (sprints), the team can prioritize important functionalities, deliver working modules incrementally, and ensure high system quality, usability, and reliability.

3.1.1 Sprint 1: Requirement Gathering and System Planning

Goals:

- Identify the core requirements and functionalities of the system.
- Define user roles and prepare the initial project roadmap.
- Finalize the technology stack and system architecture.

Activities:

- Conducted discussions with stakeholders to understand academic, Administrative, and financial challenges faced by educational institutes.
- Identified key system features such as QR-based attendance tracking, AI-powered student performance prediction, dynamic subject management, and automated parent reporting.
- Prepared a structured list of system requirements and prioritized essential features for implementation.
- Defined user stories for Admin, Manager, Teacher, and Student roles.
- Selected the technology stack including Next.js, React Native, Node.js, Express.js, PostgreSQL, and Gemini API.

Deliverables:

- Documented list of system requirements and planned features.
- Initial project timeline and development plan.
- Defined User Roles and System Requirement Specification (SRS).
- Confirmed Technology Stack and System Architecture.

3.1.2 Sprint 2: Authentication and Student Management**Goals:**

- Develop secure authentication and centralized student management functionalities.

Activities:

- Will be designed responsive registration and login interfaces for web and mobile platforms.
- Will be implemented secure authentication and role-based access control mechanisms.
- Will be developed the student management module for handling student records and profiles.
- Will be implemented subject enrollment and dynamic subject selection functionalities.

Deliverables:

- Secure Authentication Module.
- Student Management and Subject Enrollment System.

3.1.3 Sprint 3: QR Attendance and Fee Management**Goals:**

- Implement attendance tracking and automated financial management features.

Activities:

- Will be developed the QR-based attendance marking system for students.
- Will be implemented real-time attendance validation and storage.
- Will be developed subject-wise fee calculation and payment tracking functionalities.
- Will be implemented automated fee restriction mechanisms for unpaid students.

Deliverables:

- QR-Based Attendance System.
- Automated Fee and Payment Management Module.

3.1.4 Sprint 4: Examination and Academic Monitoring**Goals:**

- Develop academic monitoring and examination management functionalities.

Activities:

- Will be implemented online examination and quiz management modules.
- Will be developed marks management and academic progress tracking features.
- Will be created dashboards for viewing student performance and attendance summaries.
- Will be integrated report generation functionalities for academic monitoring.

Deliverables:

- Examination and Quiz Management System.
- Academic Monitoring and Reporting Module.

3.1.5 Sprint 5: AI Integration and Intelligent Features**Goals:**

- Integrate AI-powered functionalities into the system.

Activities:

- Will be integrated into the AI chatbot for academic and system-related assistance.
- Will be developed the AI-assisted student performance prediction module implementing the Weighted Scoring engine ($\text{attendance} \times 0.30 + \text{quiz} \times 0.20 + \text{assignment} \times 0.20 + \text{exam} \times 0.30$) and Academic Growth Trend Analysis to classify students as Excellent, Good, Average, or At-Risk, with Gemini API integration for personalized recommendations.
- Will be implemented intelligent exam recommendation functionalities based on student performance levels.
- Will be developed the automated parent reporting and notification system.

Deliverables:

- AI Chatbot Assistance System.
- Student Performance Prediction and Exam Recommendation Modules.

3.1.6 Sprint 6: Reward System, Testing, and Optimization**Goals:**

- Implement student reward functionalities and ensure system reliability.

Activities:

- Will be developed the progress point and reward management system for students.
- Will be conducted unit testing, integration testing, and User Acceptance Testing (UAT).
- Will be fixed with identified issues and optimized database queries, and API performance.
- Will be improved responsiveness and usability across web and mobile platforms.

Deliverables:

- Student Reward and Incentive System.
- Testing and Performance Optimization Report.

3.1.7 Sprint 7: Documentation and Final System Integration**Goals:**

- Finalize system integration and prepare project documentation.

Activities:

- Will be integrated all modules into the centralized platform.
- Will be prepared technical documentation and user guides.
- Will be conducted final system reviews and validations with stakeholders.
- Will be refined interfaces and workflows based on final feedback.

Deliverables:

- Fully Integrated PyramidEdu System.
- Technical Documentation and User Manuals.

3.1.8 Sprint 8: Maintenance and Future Enhancements

Goals:

- Monitor system performance and identify future improvements.

Activities:

- Will be monitored overall system functionality and collected stakeholder feedback.
- Will be identified potential future enhancements such as advanced analytics and additional AI capabilities.
- Will be performed maintenance activities and system refinements.

Deliverables:

- System Maintenance and Feedback Report.
- Future Enhancement Roadmap.

Throughout each sprint, Agile practices such as sprint planning, sprint reviews, and regular stakeholder feedback sessions were conducted to ensure continuous progress and effective collaboration. This iterative approach allowed the team to adapt to evolving institutional requirements, improve system quality through continuous feedback, and ensure that PyramidEdu effectively addresses the needs of Administrators, teachers, students, and parents.

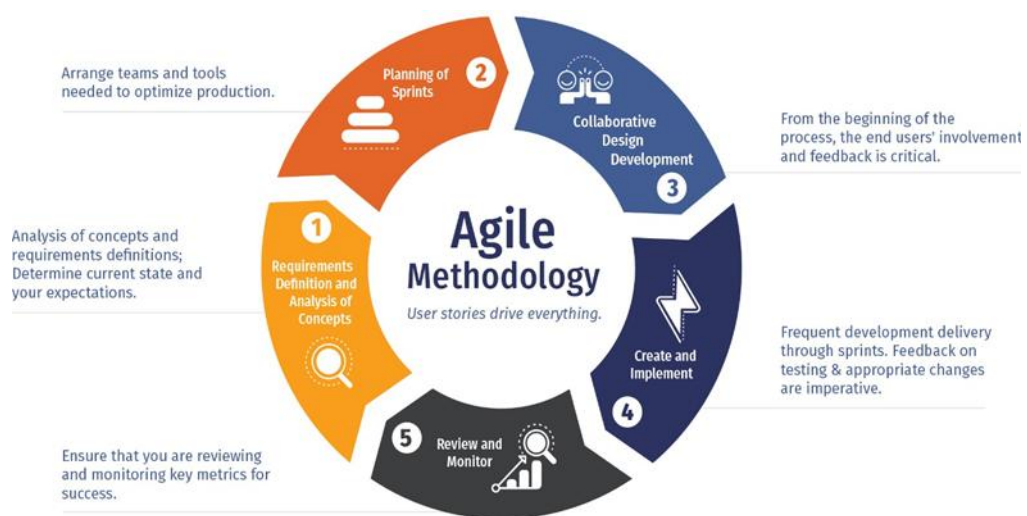


Figure 1 : Agile Methodology

4. Requirements Identification

4.1 Functional and Non-functional Requirements

4.1.1 Functional Requirements

The functional requirements for **PyramidEdu** are structured into a role-based hierarchy to ensure a secure, "System-of-Systems" architecture where each stakeholder interacts with a centralized data core.

I. Admin Functional Requirements

- **System-of-Systems Oversight:** Monitor all interconnected modules including Enrollment, Finance, and Academics via a centralized command dashboard.
- **Administrative Account Control:** Create, update, and manage Manager and Teacher accounts with full audit logs for security.
- **Financial & Salary Analytics:** Manage teacher salary records and monitor the distribution of "Free Card" scholarship rewards.
- **Global Broadcast System:** Publish institute-wide announcements and notifications that are dynamically visible to all user levels.

II. Manager Functional Requirements

- **Automated Fee Enforcement Policy:**
 - The system shall automatically monitor student payment history every month.
 - If a student fails to pay for **three consecutive months**, the system shall automatically disable the account and restrict access to all academic features.
- **Student Registry & Management:** Manage the full lifecycle of student records, from initial registration to profile updates and status changes.
- **Payment & Invoice Verification:** Review and verify student fee payments and manage digital invoices for administrative auditing.
- **QR-Based Attendance Logic:** Each student is assigned a unique permanent QR code, which is used to mark attendance by scanning each session in a secure and

efficient manner. The system also supports manual attendance marking when required.

- **Attendance & Marks Management:** Manage both attendance records (QR-based and manual) and student marks for assignments, quizzes, and examinations to ensure accurate academic tracking.
- **Automated Parent Reporting System:** Automatically generates student performance reports based on attendance, marks, and exam results, and periodically sends them to parents through digital channels (e.g., email or notifications) to ensure continuous parent involvement.

III. Teacher Functional Requirements

- **Academic Progress Logging:** Enter and update student marks for various assessments, quizzes, and examinations.
- **Digital Learning Materials:** Upload and organize notes, PDFs, and supplementary study materials for assigned subjects.
- **Performance Monitoring:** Access detailed progress reports and identifying "At-Risk" students through the system's analytical tools.
- **AI-Powered NLP Chatbot (Students & Teachers Only):**
 - An NLP-powered AI chatbot allows teachers to type questions freely. The chatbot understands intent using the Google Gemini API [\[6\]](#) and retrieves live academic data to provide intelligent responses about student performance, marks, and at-risk alerts.
 - Teachers can query student performance data, get at-risk student summaries, and receive AI-generated academic monitoring insights.

IV. Student Functional Requirements

- **Online Quiz and Examination System:**
 - Students can participate in online quizzes and exams through the platform.
 - Results are automatically stored and used for academic evaluation and progress tracking.
- **AI-Powered NLP Chatbot (Students & Teachers Only):**

- An NLP-powered AI chatbot allows students to type questions freely in natural language. The chatbot understands intent via the Google Gemini API [\[6\]](#), retrieves relevant academic information from the system, and provides intelligent responses including attendance status, marks, weak subject analysis, and study recommendations.
- Students can ask questions such as “What is my attendance percentage?”, “Show my weak subjects”, “Give me study tips for programming”, or “Summaries PDF”.
- **Performance Analytics System:**
 - Provides detailed analysis of student performance based on quizzes, exams, and attendance data.
 - Helps identify weak areas and track academic progress over time.
- **Intelligent Exam Recommendation System:** Receive personalized exam and quiz recommendations based on performance level (At-Risk, Average, or Excellent). Recommendations are generated by analyzing marks, attendance, and previous quiz results to support personalized learning and academic improvement.
- **AI-Assisted Performance Prediction:** Receive AI-generated performance predictions, trend analysis (Improving / Stable / Declining), and personalized academic recommendations on the student dashboard to track progress and take timely action.
- **Learning Materials Management:**
 - Students can access and download lecture notes, study materials, and resources uploaded by teachers.
 - Ensure centralized and organized content distribution.
- **Authentication:**
 - Secure registration and login system for students.
- **Payment Management System:**
 - Handles student fee tracking and payment status updates.
 - Automatically manages subject-wise fee calculations and payment records.

4.1.2 Non-functional Requirements

I. Performance: The system must efficiently handle concurrent access by up to 100 users without noticeable delays. Page load times should remain under 2 seconds, while the AI chatbot and QR scanning validation should process within 1 second.

II. Scalability: The system must support the effortless addition of new subjects, student batches, or payment tiers without modifying the core logic. The "System-of-Systems" architecture must allow for future module integrations (e.g., Library Management) with minimal disruption.

III. Usability: The interface must be intuitive, responsive, and mobile-friendly to ensure accessibility for students and teachers. The system should provide clear navigation, user-friendly forms, and proper validation to enhance overall user experience.

IV. Availability & Reliability: The system must ensure a minimum of 99.9% uptime. Attendance logs and payment statuses must be auto saved in real-time to prevent data loss due to unexpected network interruptions during mobile usage.

V. Security Requirements: Access to sensitive student records and financial data must be restricted via Role-Based Access Control (RBAC). All data transmissions must be encrypted via SSL/TLS, and student passwords must be stored using strong hashing algorithms (e.g., BCrypt).

4.2 System Requirements

4.2.1 Hardware Requirements

- **Development & Manager PC:** Core i5/Ryzen 5+, 8GB RAM, SSD (Used for system management and database Administration).
- **User Smartphones:** Android 10+ / iOS 13+ (Used by students for QR scanning and chatbot interaction).
- **Connectivity:** Stable 4G/5G or Wi-Fi for real-time data synchronization.

4.2.2 Software Requirements

- **Operating System:** Windows 10+ or macOS (Development and Manager tasks).
- **Web Browsers:** Modern versions of Chrome, Edge, or Brave.
- **Frontend Framework:** Next.js 14+ and React Native.
- **Backend Runtime:** Node.js (Express.js).
- **Database:** PostgreSQL (Relational data storage).
- **AI Integration:** Gemini API (For NLP chatbot, PDF summarization, AI question generation, academic recommendations, and prediction intervention alerts).
- **Version Control:** Git and GitHub for source code management.
- **Postman:** Used for testing, debugging, and validating REST APIs during backend development.

4.2.3 Technology

To fulfill the complex functional requirements and the "System-of-Systems" architecture, **PyramidEdu** utilizes a modern, scalable tech stack. This choice ensures high performance for real-time operations (like QR scanning) and robust data integrity for financial management.

I. Frontend Technologies: Next.js, React Native, HTML5, CSS3, Tailwind CSS

- **Purpose:** Used to develop responsive web and mobile user interfaces.
- **Role:**
 - Next.js provides a high-performance React-based web application with Server-Side Rendering (SSR).
 - React Native enables cross-platform mobile application development for Android and iOS devices.
 - HTML5, CSS3, and Tailwind CSS are used to structure, style, and develop responsive and modern user interfaces efficiently.

II. Backend Technologies: Node.js & Express.js

- **Purpose:** Handles server-side logic, REST APIs, authentication, and system integrations.

- **Role:** Acts as the core processing layer of the system by managing QR validation, fee restriction logic, user authentication, notifications, and data communication between modules.

III. Database: PostgreSQL

- **Purpose:** A robust, open-source relational database system designed for data integrity, scalability, and complex queries.
- **Role:** Stores critical information including student profiles, transaction histories, multi-subject enrollment data, and granular attendance logs.

IV. AI and Analytics Technologies: Google Gemini API & TypeScript Analytics Engine

- **Purpose:** Provides AI-powered prediction, recommendation, and chatbot functionalities.
- **Role:**
 - A TypeScript-based Weighted Scoring and Trend Analysis engine processes student attendance, quiz results, and examination scores to detect academic growth patterns and classify students as improving, stable, or at-risk. Gemini API [\[6\]](#) powers the AI chatbot [\[9\]](#) and generates intelligent personalized academic recommendations and intervention responses.
 - The Weighted Scoring engine operates entirely within the Node.js backend using TypeScript, requiring no external ML services.

V. Mobile Capabilities: React Native / Progressive Web Application (PWA)

- **Purpose:** Enables mobile accessibility and cross-device compatibility [\[10\]](#).
- **Role:** Provides students with mobile-based QR attendance, notifications, chatbot access, subject management, and academic monitoring features.

VI. Programming Languages

- **TypeScript:** Used for frontend, backend, and mobile application development.
- **SQL:** Used for database queries and relational data management.

- **JavaScript/TypeScript (Backend):** Used for analytics engine development, weighted scoring logic, and AI-based trend detection processing within the Node.js backend

VII. ORM Framework: Prisma ORM

- Prisma ORM is used to efficiently manage and simplify database interactions between the Node.js backend and the PostgreSQL database.

4.3 User Roles

I. Admin (Institute Director)

The admin is the highest authority in the system who manages overall academic structure, system configuration, and institutional control.

- Create and manage institutes, departments, and academic structure.
- Manage Managers, Teachers, Students, and system-wide user roles.
- Configure system settings including fees, rules, grading rules, and policies.
- Monitor overall system analytics (student performance, attendance, usage).
- Approve or restrict access to institutional modules.
- Manage AI-based features such as prediction rules and report generation.
- Ensure system security, data backup, and access control.

II. Manager (Institute Management)

The Manager manages day-to-day institutional operations and ensures smooth academic and administrative workflows.

- Manage student registrations and academic records.
- Create and manage subjects and subject allocations.
- Mark student attendance (QR-based system integration).
- Monitor attendance, examinations, and academic schedules.
- Generate and manage student fee records.
- Handle notifications and communication to students and parents.
- View student performance reports and analytics dashboards.
- Manage exam schedules, quizzes, and result publication.
- Oversee AI-generated reports and system recommendations.

III. Teacher

Teachers are responsible for academic delivery, student evaluation, and performance tracking.

- Manage assigned subjects and teaching schedules.
- Conduct exams, quizzes, and assignments.
- Enter and update student marks and assessments.
- Review student performance and learning progress.
- Provide feedback and academic guidance to students.
- Communicate academic updates through the system.
 - Use the NLP AI chatbot to query student performance data, identify at-risk students, get AI-generated academic monitoring insights, and summarize uploaded study materials.

IV. Student

Students are the primary users of the system who access academic services and track their progress.

- Enroll in subjects and dynamically select or remove subjects.
- View attendance records and academic progress.
- Participate in exams, quizzes, and assignments.
- Receive performance reports and AI-based predictions.
- Access study materials and notifications.
 - Use the NLP AI chatbot to ask academic questions freely, check attendance and marks, get weak subject analysis, receive personalized study guidance, summarize uploaded PDFs, and generate revision questions from study materials.
- View fee status and payment history.
- Communicate with teachers through the system.
- Earn progress points and performance-based rewards.

5. Project Plan (Gantt chart)

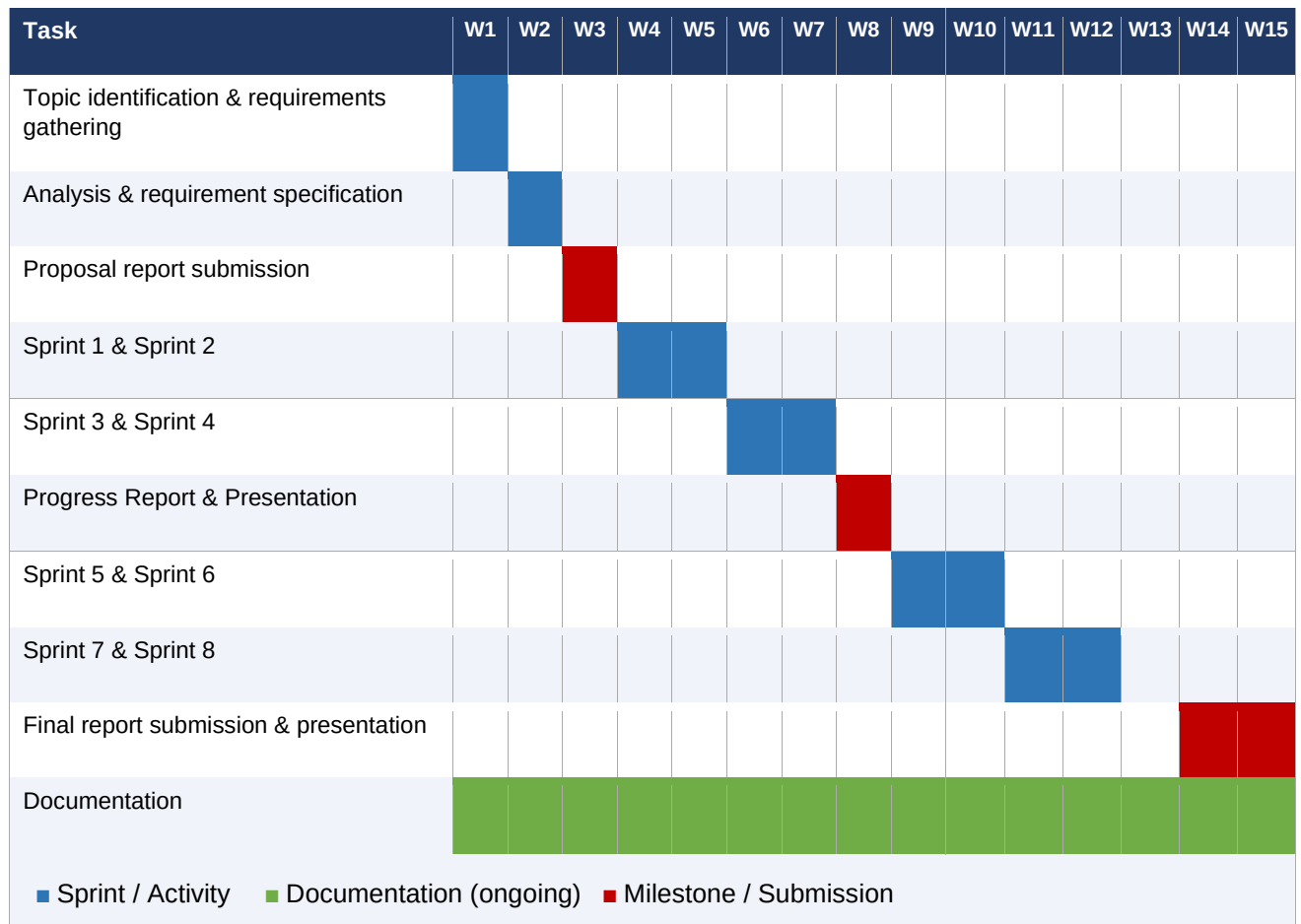


Figure 2 : Project plan (Gantt chart)

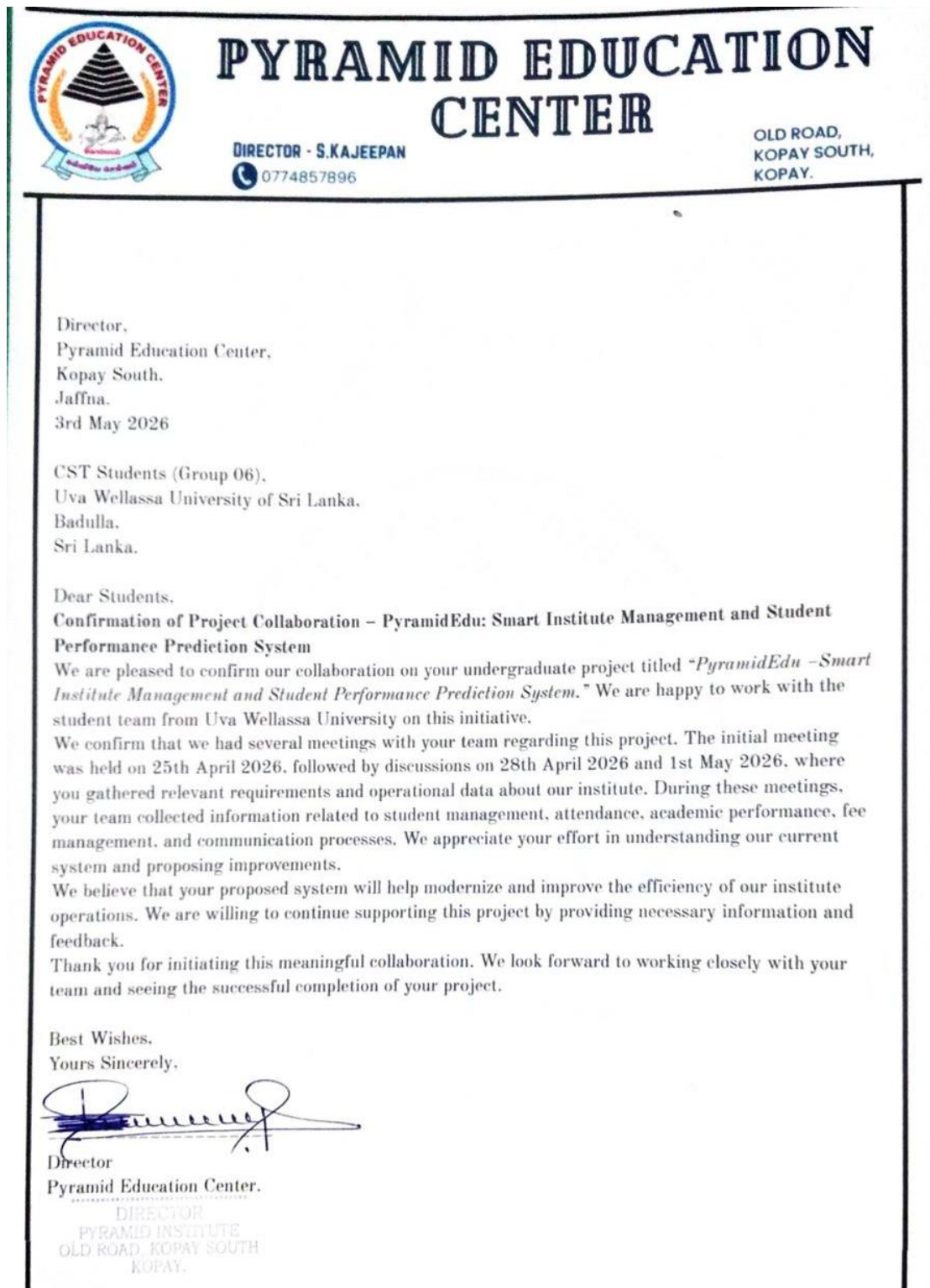
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7. Appendices



Appendix A : Client Confirmation Letter